

6.5 Forecasting ^{A↓}

□ Guiding Questions

1. What is forecasting, and why is it useful for financial decision-making?
2. How can you identify trend, seasonality, cycles, and noise in a time series?
3. Which forecasting method should you use for a stable, trending, or seasonal series?
4. What are the strengths and limitations of simple moving averages, exponential smoothing, regression forecasts, and seasonal decomposition?
5. How do you evaluate whether one forecast is better than another?
6. How should forecast uncertainty be communicated to non-technical stakeholders?



EXPLORE

What you will learn

- How to recognise common patterns in time series data, including trend, cycle, seasonality, and irregular shocks.
- How to choose a forecasting method that matches the structure of the data.
- How to create simple forecasts in R using moving averages, exponential smoothing, regression, and seasonal decomposition.
- How to interpret forecast outputs, including fitted values, future predictions, and confidence intervals.
- How to compare forecast accuracy.
- How to communicate uncertainty clearly, rather than presenting forecasts as exact predictions.

What is Forecasting?

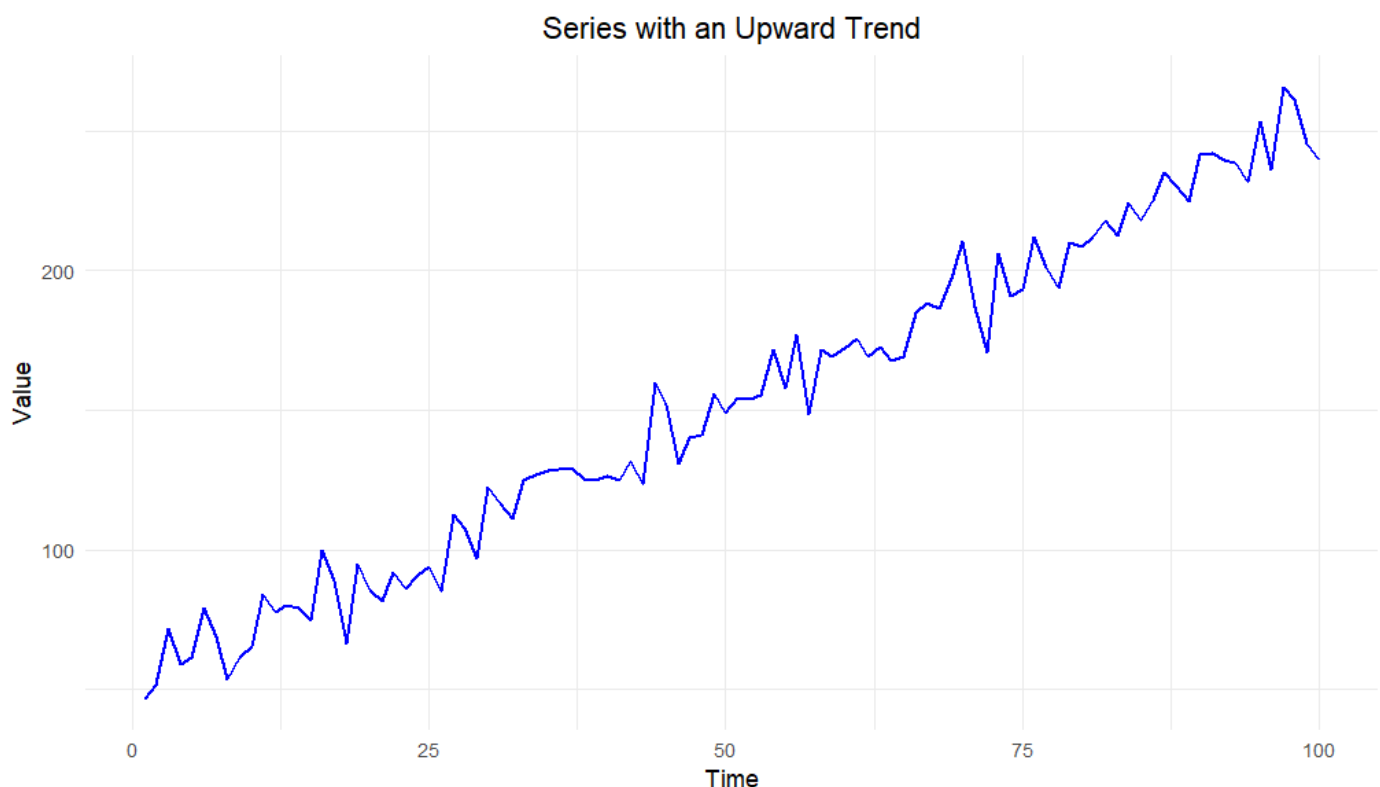
Forecasting is the process of using historical data to make informed predictions about future values. In financial analytics, forecasting is used to project revenues, estimate costs, anticipate demand, and plan for uncertainty. A good forecast does not eliminate uncertainty; it structures it, giving decision-makers a disciplined basis for planning rather than guesswork.

Every forecasting method makes assumptions about the structure of the data. Before choosing a method, an analyst must understand what patterns are present in the time series.

Trends, Cycles, and Seasonality

Trend

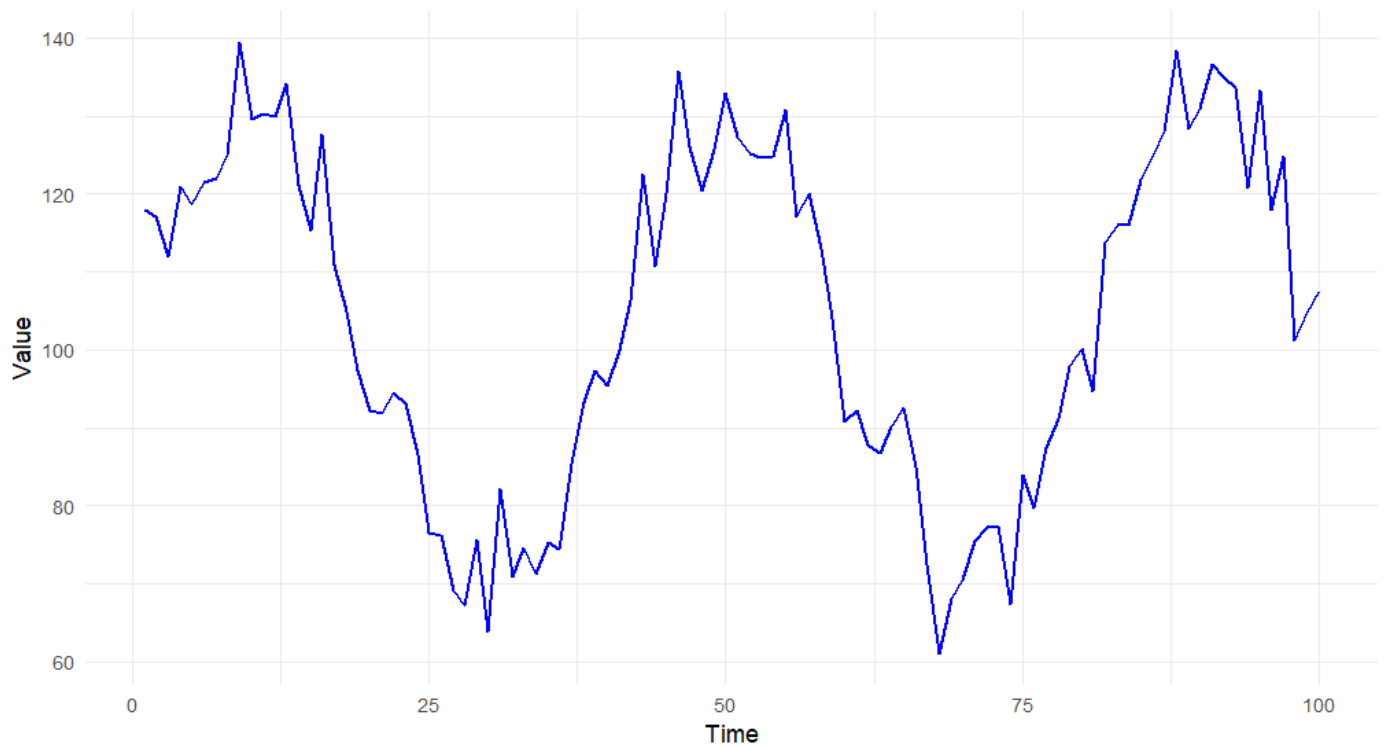
A trend is a long-run upward or downward movement in the data over time. Revenue that grows steadily year after year exhibits an upward trend. A company's workforce that shrinks over a decade exhibits a downward trend. Trends are persistent and directional. The graph below shows a trending series; the series generally moves upward over time, even though there is some random variation.



Cycle

A cycle is a wave-like fluctuation around the trend that repeats over a period longer than one year. Business cycles (expansions followed by contractions) are the most familiar example. Unlike seasons, cycles are irregular in both length and amplitude. They are difficult to forecast precisely because their timing and depth depend on complex macroeconomic forces that are not fully predictable. The series below is cyclical because it rises and falls in waves over a longer period, rather than repeating at a fixed short seasonal interval.

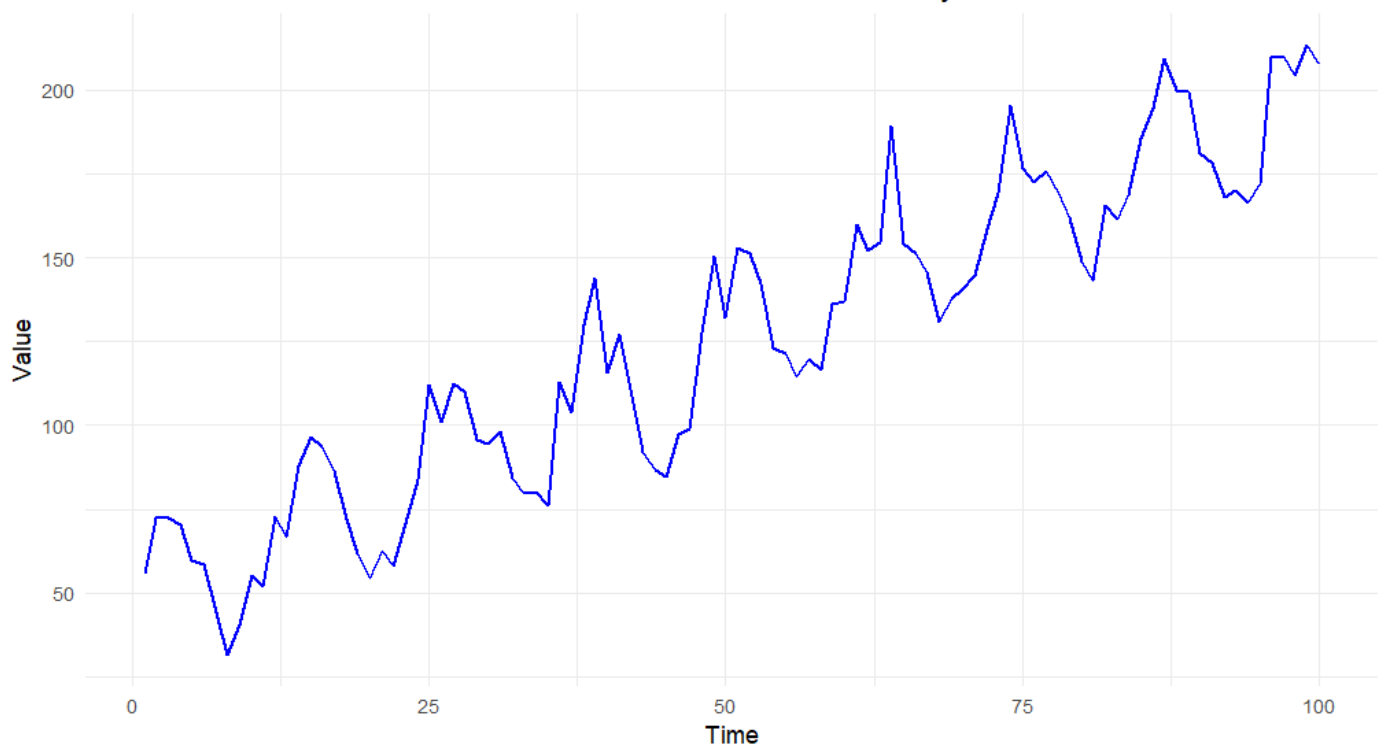
Series with Cycles



Seasonality

Seasonality is a pattern that repeats at fixed, known intervals within a year. Retail sales spike every December. Hotel bookings peak every summer. Ice cream sales rise every quarter two and three. Seasonal patterns are regular and predictable, which makes them relatively straightforward to model once identified. The series below has a trend because it rises over time, and seasonality because the pattern repeats at regular intervals.

Series with Trend and Seasonality



Irregular or Noise

The irregular component is what remains after trend, cycle, and seasonality have been accounted for. It represents random shocks, one-off events, and measurement error. It is by definition unpredictable and should not be modelled directly.

A Forecasting Problem: The Business Context

Imagine you are a financial analyst working for a major Australian airline. Your job is to forecast passenger demand because everything else in the business depends on it. Fleet scheduling, crew rostering, fuel purchasing, and revenue projections all flow from a demand forecast. An accurate forecast allows the airline to match capacity to demand efficiently, avoiding the twin costs of flying half-empty aircraft and turning away passengers on overbooked routes.

Before building any model, you need to understand the structure of the data. Does demand grow over time or fluctuate around a stable level? Are there predictable seasonal patterns, such as school holidays and the Christmas peak, that repeat every year? The answers determine which forecasting method is appropriate. Applying the wrong method to the wrong type of data produces forecasts that are confidently wrong, which is worse than acknowledging uncertainty honestly.

The dataset used in this module records monthly international airline passenger numbers from 1949 to 1960. The analytical problem it represents is live and consequential in every airline, retailer, hotel group, and utility company that plans ahead. As you work through the code, think of yourself as an analyst preparing a demand forecast to present to the operations and finance teams next week.

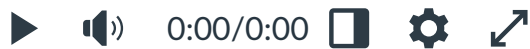
Watch

[Forecasting: Simple Moving Average and Exponential Smoothing](#)



Details

Transcript



[R Script \(https://learn.usc.edu.au/courses/35359/files/3226180?wrap=1\)](https://learn.usc.edu.au/courses/35359/files/3226180?wrap=1) 
(https://learn.usc.edu.au/courses/35359/files/3226180/download?download_frd=1)

First Step: Plot the Data

```
install.packages("forecast")

library(forecast)

library(ggplot2)

# Convert to time series object

# start = c(1949, 1) means January 1949

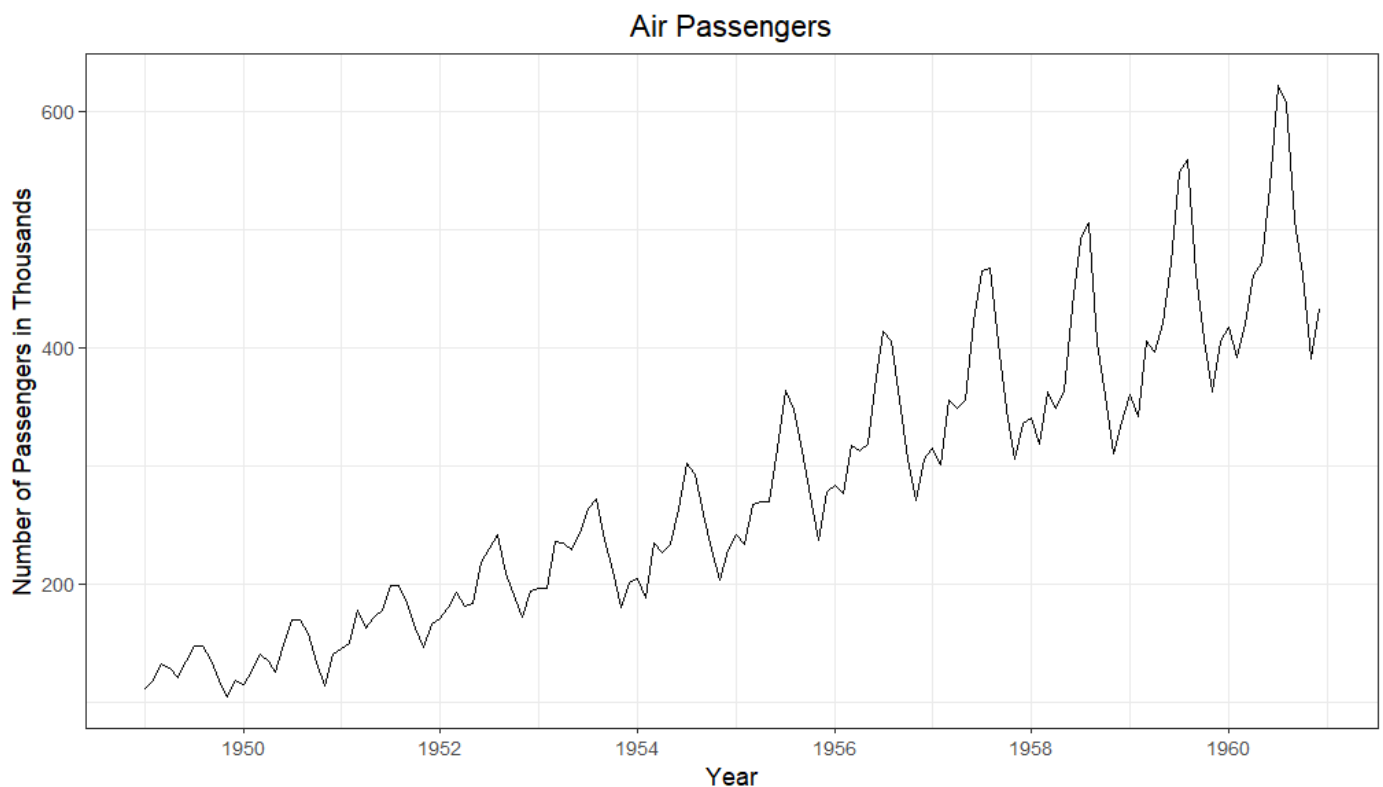
# frequency = 12 means monthly data

passengers <- ts(AirPassengers, start = c(1949, 1), frequency = 12)

# now plot the data

autoplot(passengers) +
  labs(
    title = "Air Passengers",
    x = "Year",
    y = "Number of Passengers in Thousands"
  ) +
  theme_bw(base_size = 15) +
```

```
theme(
  plot.title = element_text(hjust = 0.5)
)
```



Simple Moving Average

A simple moving average forecasts the next value by averaging the most recent k observations. It works well for stationary series where there is no trend and no seasonality. The k periods are averaged equally — older and newer observations receive the same weight.

The key decision is choosing k . A small k makes the forecast sensitive to recent changes but noisy. A large k produces a smoother forecast but reacts slowly to genuine shifts in the data.

When to use it: Short-term forecasting of stable series. Useful as a benchmark against which more sophisticated methods are compared.

When not to use it: Trending or seasonal series. The moving average will always lag behind a trend and will not capture repeating seasonal patterns.

The forecast for period $t + 1$ is the arithmetic mean of the most recent k observations:

$$\hat{y}_{t+1} = \frac{1}{k} \sum_{i=0}^{k-1} y_{t-i} = \frac{y_t + y_{t-1} + y_{t-2} + \cdots + y_{t-k+1}}{k}$$

Where k is the number of periods included in the average. For example, with $k = 3$ and observations of 100, 110, and 120, the forecast for the next period is $(100 + 110 + 120) / 3 = 110$.

How this is implemented in R

- Fit a simple moving average with order 12
- order = 12 averages the last 12 months (one full year)

```
ma_model <- ma(passengers, order = 12)
```

```
# Forecast 12 months ahead
```

```
ma_forecast <- forecast(ma_model, h = 12)
```

```
# Plot the forecast
```

```
# autoplot() is a ggplot2-compatible plot for forecast objects
```

```
# It automatically adds confidence intervals
```

```
autoplot(ma_forecast) +
```

```
labs(
```

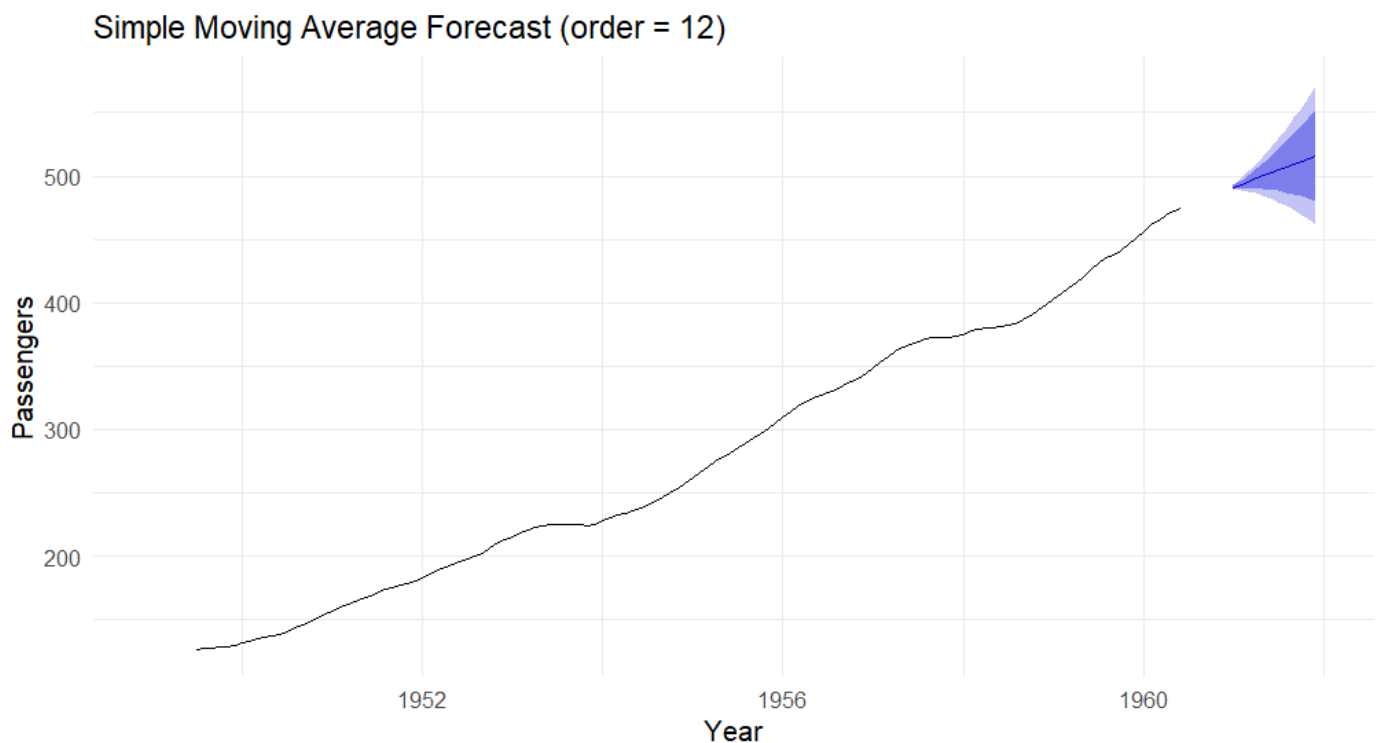
```
  title = "Simple Moving Average Forecast (order = 12)",
```

```
  x = "Year",
```

```
  y = "Passengers" ) +
```

```
  theme_minimal(base_size = 13)
```

```
print(ma_forecast)
```



Notice that the moving average smooths out month-to-month noise but it cannot follow the upward trend in the data; the forecast is a little flatter than what the trend might suggest. The

forecast will systematically underpredict future values because it averages in older, lower observations. This illustrates why moving averages are best reserved for stationary series.

Exponential Smoothing

Exponential smoothing improves on the simple moving average by giving more weight to recent observations and less weight to older ones. The rate at which weights decay is controlled by a smoothing parameter alpha, which ranges from 0 to 1. A high alpha means recent observations dominate. A low alpha means older observations still carry substantial weight.

The `ets()` function in the forecast package selects the best variant of exponential smoothing automatically by minimising a model selection criterion. The three main variants are:

Simple exponential smoothing — for stationary series with no trend and no seasonality.

Holt's linear trend method — adds a trend component, making it suitable for series that trend upward or downward without seasonality.

Holt-Winters method — adds both trend and seasonality, making it the most flexible and appropriate for series with both features.

In practice, `ets()` chooses among these variants for you. You can inspect the chosen model by printing the result.

When to use it: Trending or stationary series. Holt-Winters handles seasonality as well, making `ets()` one of the most versatile methods available.

When not to use it: Very short series where there are not enough observations to estimate the smoothing parameters reliably.

Simple Exponential Smoothing (no trend, no seasonality)

The forecast is a weighted average of the current observation and the previous forecast:

Forecast for next period = $\alpha \times \text{current observation} + (1 - \alpha) \times \text{previous forecast}$

where α is between 0 and 1.

A high α gives more weight to recent observations. A low α gives more weight to older history.

Written out over time, the forecast gives the highest weight to the most recent observation, then smaller and smaller weights to older observations:

Forecast = $\alpha \times \text{current value} + \alpha(1 - \alpha) \times \text{previous value} + \alpha(1 - \alpha)^2 \times \text{value two periods ago} + \dots$

This is why it is called exponential smoothing: the weights decline exponentially as observations get older.

How This is Implemented in R

```
ets_model <- ets(passengers) # use the ets function; this function automatically selects alpha
through an optimisation process
```

```
# See which model was selected and its parameters; you can see the chosen alpha using the
summary function
```

```
summary(ets_model)
```

```
# Forecast 12 months ahead; set h = 12
```

```
ets_forecast <- forecast(ets_model, h = 12)
```

```
# Plot the forecast with confidence intervals
```

```
autoplot(ets_forecast) +
```

```
  labs(
```

```
    title = "Exponential Smoothing Forecast (ETS)",
```

```
    x     = "Year",
```

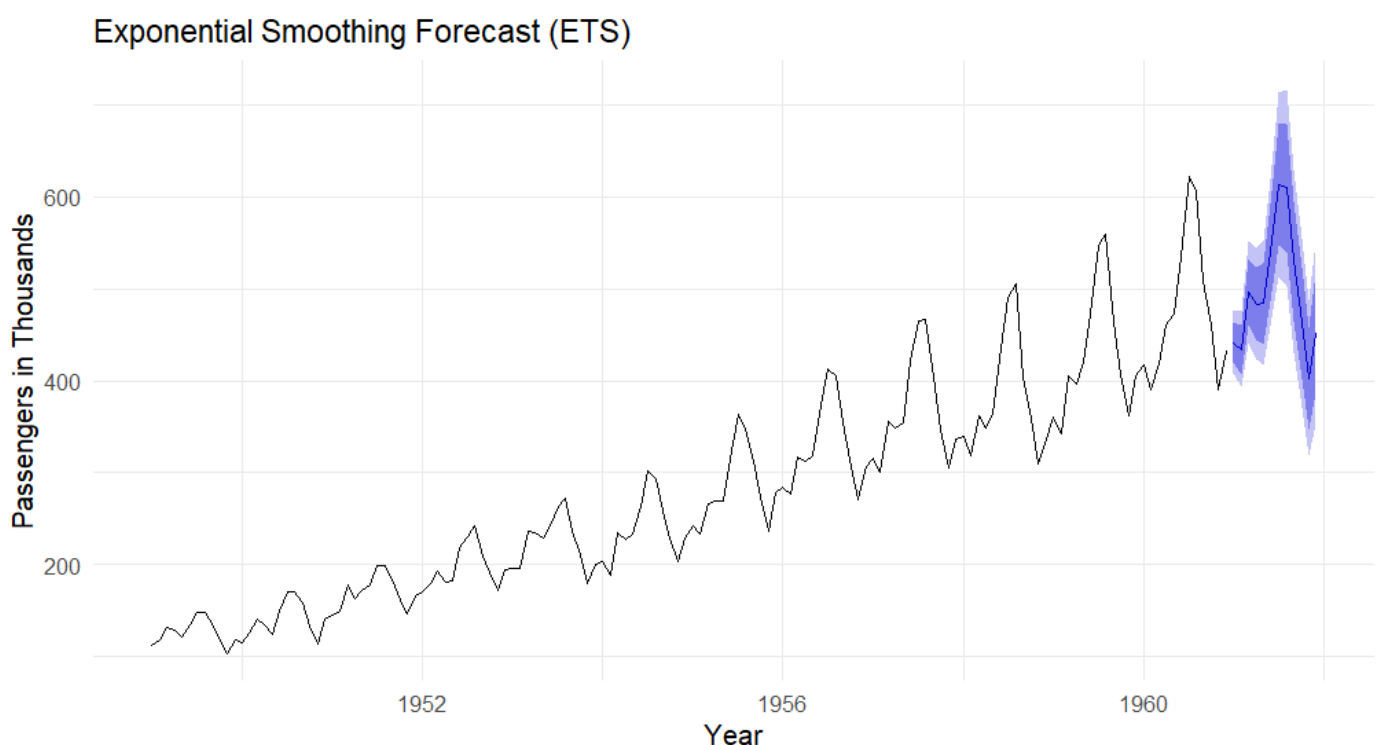
```
    y     = "Passengers in Thousands"
```

```
  ) +
```

```
  theme_minimal(base_size = 13)
```

```
# Print the forecast values and confidence intervals
```

```
print(ets_forecast)
```



Notice that the forecast will capture both the upward trend and the seasonal peaks and troughs, which the simple moving average could not. The `summary()` output shows the smoothing parameters alpha, beta, and gamma, which control how quickly the model adapts to changes in level, trend, and seasonality respectively.



Watch

Forecasting: Linear Regressions

Linear Regression Trend Forecast

Linear regression fits a straight line through the historical data, where time is the predictor variable. This makes it suitable for series with a clear linear trend. The fitted line is then extended forward to produce the forecast.

The advantage of regression over smoothing methods is interpretability. The slope coefficient tells you directly how much the series increases per period on average. The disadvantage is rigidity, as a straight line cannot capture curvature, and if the trend is accelerating or decelerating, the regression forecast will be systematically biased.

Seasonal patterns can be incorporated by adding dummy variables for each month or quarter. This allows the regression to capture both the linear trend and the repeating seasonal pattern simultaneously.

When to use it: Series with a stable linear trend. Series where interpretability of the forecast is important.

When not to use it: Series with a non-linear trend. Series where the relationship between time and the outcome is not approximately linear.

Linear Regression Trend Forecast

Trend only

A regression trend model explains the outcome as a straight-line function of time:

Outcome = intercept + trend effect × time + error

In simple terms:

Predicted value = starting level + average change per period × time period

The intercept is the starting level of the series. The trend effect shows how much the outcome changes, on average, each period. The time index is simply the order of the observations, such as 1, 2, 3, and so on. The error term captures the part of the outcome that the model does not explain.

To forecast a future period, we extend the fitted line forward:

Forecast = estimated intercept + estimated trend effect × future time period

How This is Implemented in R

```
# tslm() fits a linear regression model to a time series
regression_model <- tslm(passengers ~ trend)

summary(regression_model)

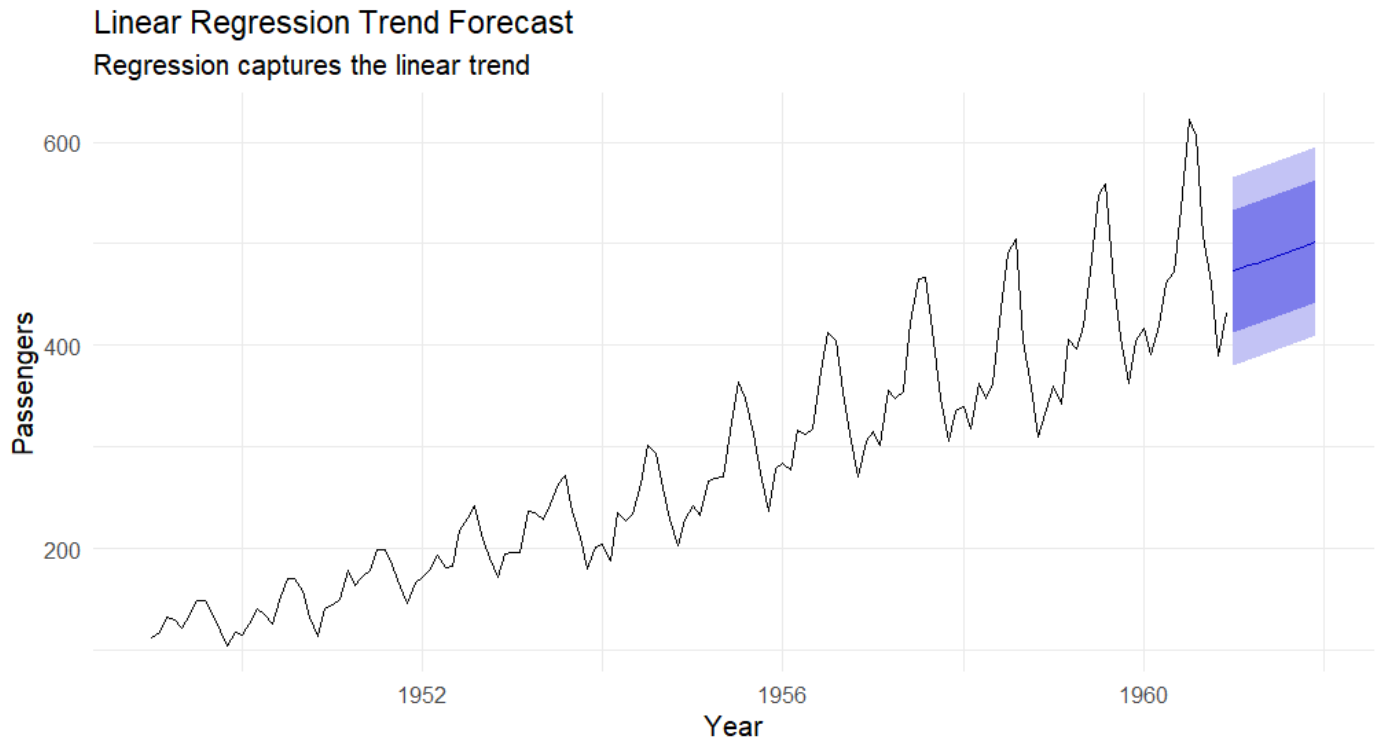
# Forecast 12 months ahead
# h = 12 produces forecasts for the next 12 periods
regression_forecast <- forecast(regression_model, h = 12)

# Plot the forecast
autoplot(regression_forecast) +
  labs(
    title = "Linear Regression Trend Forecast",
    subtitle = "Regression captures the linear trend",
    x = "Year",
    y = "Passengers"
  ) +
  theme_minimal(base_size = 13)

# Print the forecast
print(regression_forecast)
```

```
# Compare accuracy with the ETS model
```

```
accuracy(regression_forecast)
```



Notice that the seasonality is not captured. However, the upward trend is captured.

Linear Regression Trend and Seasonality Forecast

If the data have both a trend and a repeating seasonal pattern, we add seasonal indicators to the regression model.

For monthly data, this means adding month variables. One month is left out as the comparison month, often January. The remaining months are compared against January.

In simple terms:

Outcome = intercept + trend effect × time + seasonal month effects + error

The trend effect captures the average movement over time. The seasonal month effects capture whether each month is usually higher or lower than the base month, after accounting for the trend.

For example, if January is the base month, the February effect tells us how much higher or lower February tends to be compared with January, holding the overall trend constant.

Use the same code as above. The only change required is the addition of season in the `tslm` function.

```
regression_model_season <- tslm(passengers ~ trend + season)
```

```
# View the regression output
```

```
summary(regression_model_season)
```

```
# Forecast 12 months ahead
```

```
# h = 12 produces forecasts for the next 12 periods
```

```
regression_forecast_season <- forecast(regression_model_season, h = 12)
```

```
# Plot the forecast
```

```
autoplot(regression_forecast_season) +
```

```
labs(
```

```
  title = "Linear Regression Trend and Seasonal Forecast",
```

```
  subtitle = "Regression captures the linear trend and monthly seasonal pattern",
```

```
  x = "Year",
```

```
  y = "Passengers"
```

```
) +
```

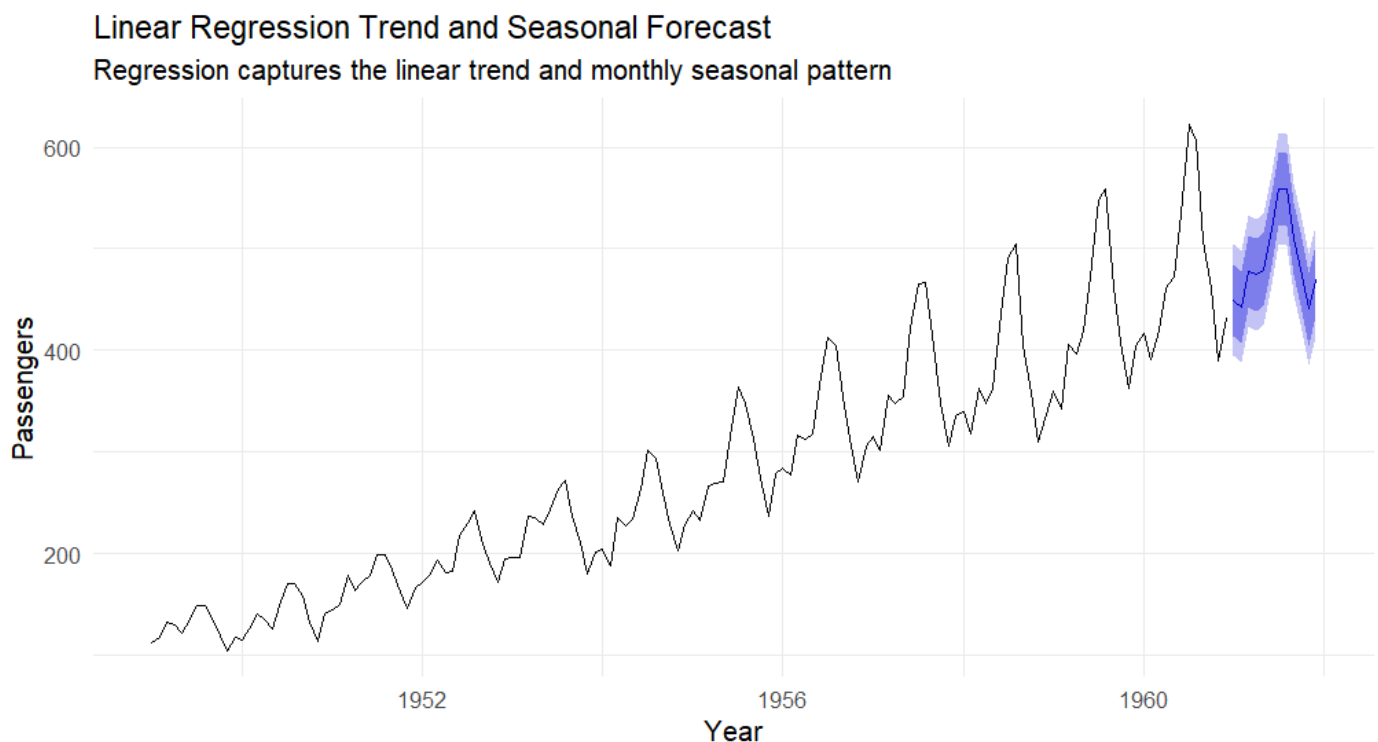
```
theme_minimal(base_size = 13)
```

```
# Print the forecast
```

```
print(regression_forecast)
```

```
# Compare accuracy with the ETS model
```

```
accuracy(regression_forecast)
```



Notice that this forecast captures the trend and the seasonality.

Comparing Forecasting Methods

No single method is best in all situations. The right method depends on the structure of the data and the forecasting horizon. The table below summarises the key characteristics of the four methods covered in this module.

Method	Strengths	Weaknesses
Simple Moving Average	Simple, transparent, good benchmark	Cannot handle trend or seasonality
Exponential Smoothing	Flexible, automatic model selection, handles trend and seasonality	Less interpretable than regression
Linear Regression	Interpretable coefficients, handles trend and seasonality with dummies	Assumes linear trend, inflexible

Evaluating forecast accuracy

The `accuracy()` function returns several error measures. The two most commonly used are:

RMSE (Root Mean Squared Error) — the square root of the average squared difference between the forecast and the actual value. Lower is better. RMSE penalises large errors more heavily than small ones.

MAPE (Mean Absolute Percentage Error) — the average percentage difference between the forecast and the actual value. Lower is better. MAPE is useful because it is expressed as a percentage, making it easy to compare across series with different scales.

When comparing models, always evaluate accuracy on a hold-out period, i.e., data the model has not seen, rather than on the training data. A model that fits the historical data perfectly but forecasts poorly is overfitted and of no practical use.

Communicating Forecast Results to Stakeholders

As a financial analyst, producing a forecast is only half the job. The other half is communicating it clearly to people who will make decisions based on it, many of whom will never look at the underlying model. The operations team needs to know whether to schedule more aircraft on the December routes. The finance team needs a revenue projection with a realistic range, not a single optimistic number. The board needs confidence that the forecast is grounded in evidence and that the risks have been considered.

The key is to tailor the communication to the decision at hand. For operational stakeholders, lead with the direction and magnitude of the forecast. Passenger demand is projected to peak in December at approximately 450,000 passengers, roughly 20 percent above the monthly average. That is the number they need to plan around. For financial stakeholders, the confidence interval matters as much as the point forecast. Present a base case, a downside, and an upside, and be explicit about what would have to be true for each to materialise. For senior leadership, connect the forecast to a decision. If demand grows as projected, the airline will need two additional aircraft on the Sydney to Melbourne route by the third quarter. If the downside scenario materialises, that decision can be deferred.

One practical rule is worth remembering. A forecast presented as a single precise number invites false confidence. A forecast presented with an honest range, a clear statement of the assumptions behind it, and an explicit discussion of what could go wrong is far more useful to a decision-maker, and far more defensible when reality turns out to be different from the projection.